

WEEK 4 Day 1

Seaborn in Python

Seaborn is a Python library built on top of Matplotlib that is used for creating attractive and informative statistical visualizations. It provides a high-level interface for drawing beautiful graphs with less code.

Seaborn works especially well with Pandas DataFrames, making it a popular library in Data Analytics, Data Science, and Machine Learning.

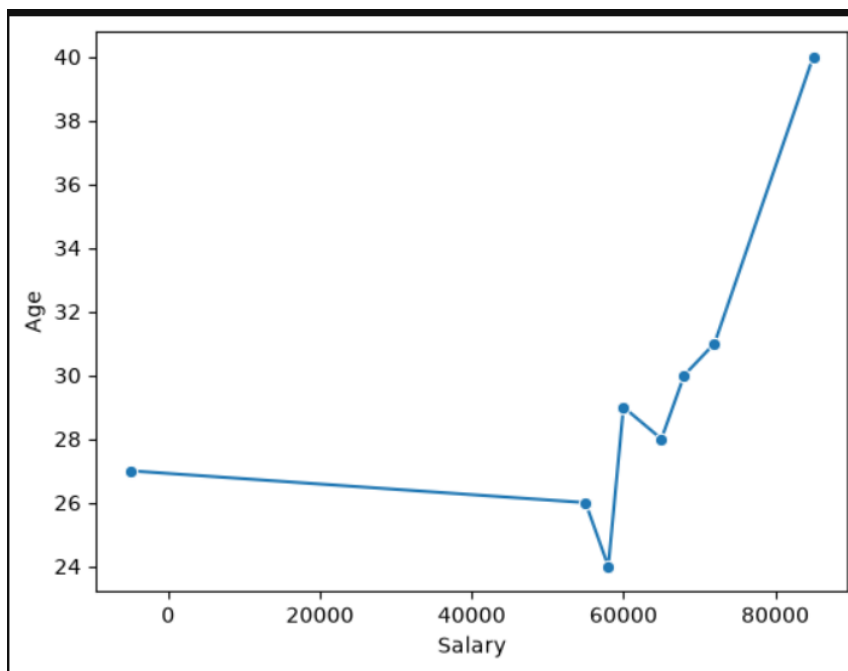
1. Line Plot

A **line plot** is used to show trends or changes in data.

```
# Line Plot
import seaborn as sns
import pandas as pd
import matplotlib.pyplot as plt

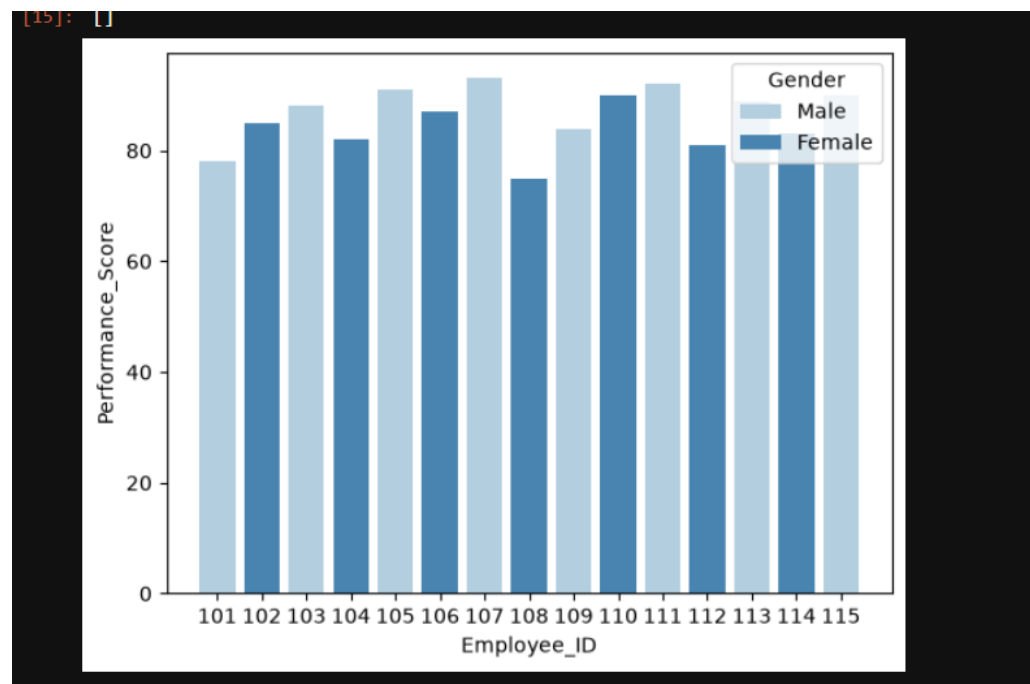
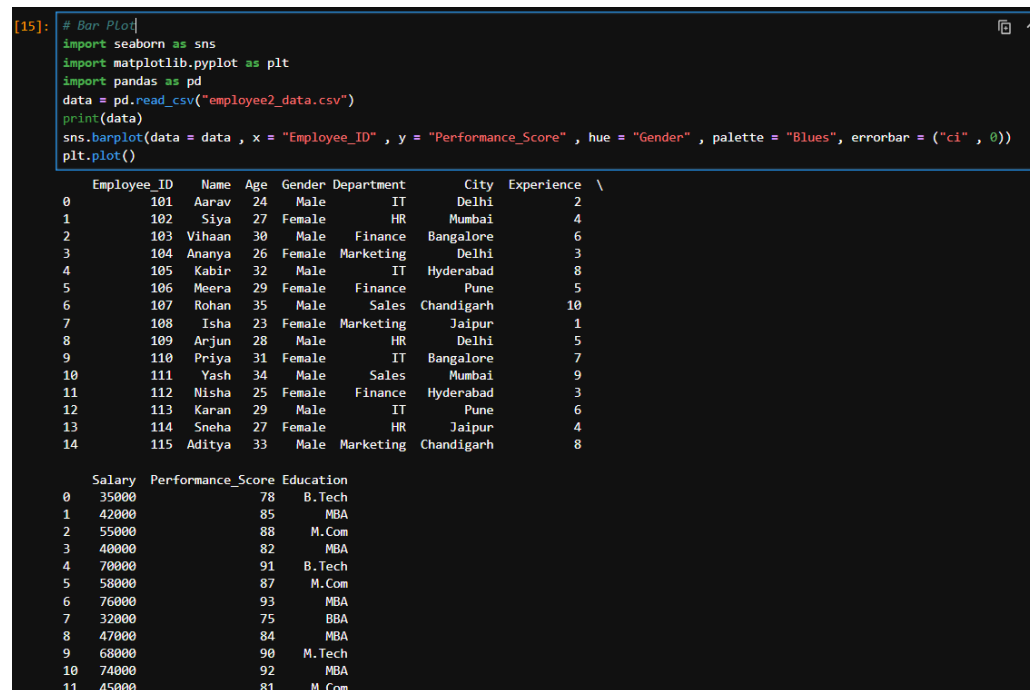
df = pd.read_csv("employee_data.csv")

sns.lineplot(
    data=df,
    x="Salary",
    y="Age",
    marker="o"
)
plt.show()
```



2. Bar Plot

A **bar plot** compares values across different categories.



3. Histogram

A **histogram** displays the frequency distribution of continuous data.

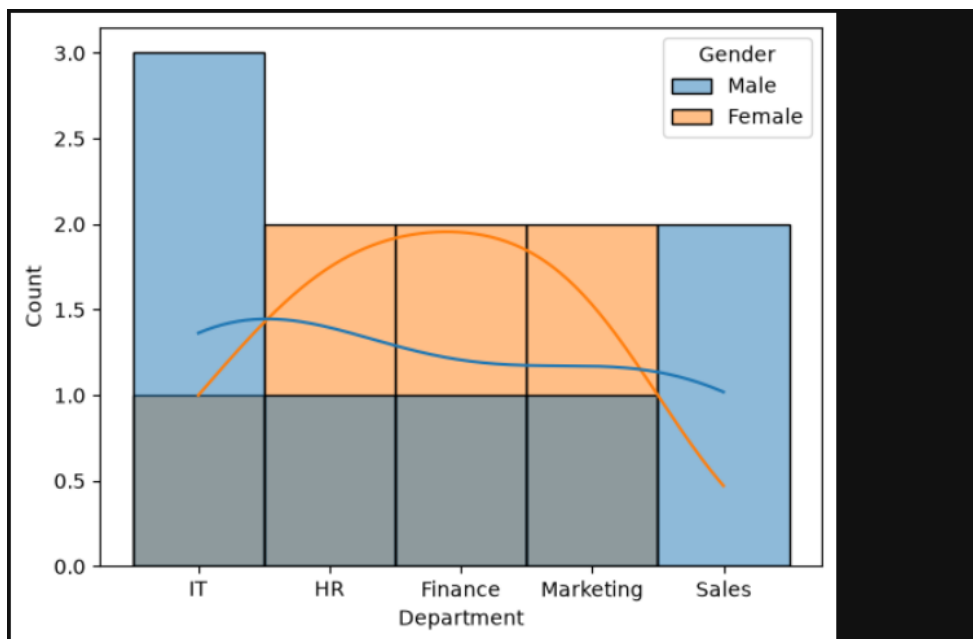
```
[26]: # Histogram
import seaborn as sns
import pandas as pd
import matplotlib.pyplot as plt

data = pd.read_csv("employee2_data.csv")
print(data)

sns.histplot(data, x = "Department", hue = "Gender", kde = True, discrete = True, bins = 25)
plt.show()
```

	Employee_ID	Name	Age	Gender	Department	City	Experience	\
0	101	Aarav	24	Male	IT	Delhi	2	
1	102	Siya	27	Female	HR	Mumbai	4	
2	103	Vihaan	30	Male	Finance	Bangalore	6	
3	104	Ananya	26	Female	Marketing	Delhi	3	
4	105	Kabir	32	Male	IT	Hyderabad	8	
5	106	Meera	29	Female	Finance	Pune	5	
6	107	Rohan	35	Male	Sales	Chandigarh	10	
7	108	Isha	23	Female	Marketing	Jaipur	1	
8	109	Arjun	28	Male	HR	Delhi	5	
9	110	Priya	31	Female	IT	Bangalore	7	
10	111	Yash	34	Male	Sales	Mumbai	9	
11	112	Nisha	25	Female	Finance	Hyderabad	3	
12	113	Karan	29	Male	IT	Pune	6	
13	114	Sneha	27	Female	HR	Jaipur	4	
14	115	Aditya	33	Male	Marketing	Chandigarh	8	

	Salary	Performance_Score	Education
0	35000	78	B.Tech
1	42000	85	MBA
2	55000	88	M.Com
3	40000	82	MBA
4	70000	91	B.Tech
5	58000	87	M.Com
6	76000	93	MBA



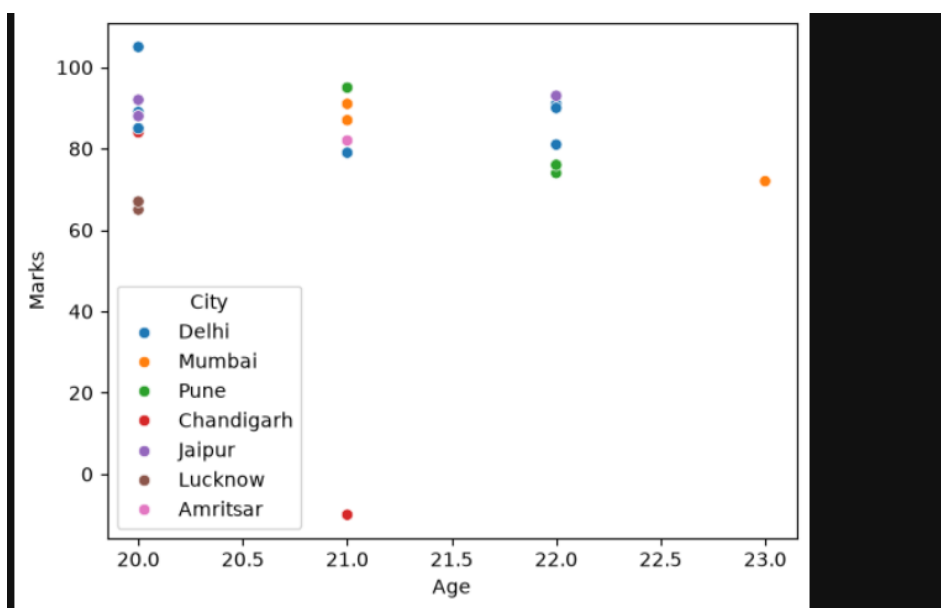
4. Scatter Plot

A **scatter plot** shows the relationship between two numerical variables.

```
[7]: # Scatter Plot

import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd
data = pd.read_csv("student_data.csv")
print(data)
sns.scatterplot(data = data , x = "Age" , y = "Marks" , hue = "City")
plt.show()
```

	Student_ID	Name	Gender	Age	Department	Semester	Marks	Attendance
0	101	Aarav	Male	20.0	CSE	5	85.0	92.0
1	102	Priya	Female	21.0	IT	5	91.0	95.0
2	103	Rahul	Male	22.0	ECE	6	76.0	88.0
3	104	Anjali	Female	20.0	CSE	5	89.0	91.0
4	105	Karan	Male	21.0	IT	5	NaN	85.0
5	106	Neha	Female	22.0	ECE	6	93.0	NaN
6	107	Rohit	Male	20.0	CSE	5	67.0	75.0
7	108	Simran	Female	21.0	IT	5	82.0	89.0
8	109	Aditya	Male	NaN	CSE	6	90.0	94.0
9	110	Kavya	Female	20.0	ECE	5	88.0	90.0
10	111	Aman	Male	23.0	CSE	7	72.0	70.0
11	112	Riya	Female	21.0	IT	5	95.0	98.0
12	113	Vikram	Male	22.0	ECE	6	81.0	87.0
13	114	Mehak	Female	20.0	CSE	5	84.0	92.0
14	115	Anjun	Male	21.0	IT	5	79.0	85.0
15	116	Nisha	Female	22.0	CSE	6	91.0	96.0
16	117	Rohan	Male	20.0	CSE	5	65.0	68.0
17	118	Pooja	Female	21.0	ECE	5	87.0	90.0
18	119	Deepak	Male	22.0	IT	6	74.0	80.0
19	120	Sneha	Female	20.0	CSE	5	92.0	97.0
20	121	Aarav	Male	20.0	CSE	5	85.0	92.0
21	122	Priya	Female	21.0	IT	5	91.0	95.0
22	123	rahul	Male	22.0	ECE	6	76.0	88.0
23	124	Anjali	Female	20.0	CSE	5	105.0	91.0
24	125	Karan	Male	21.0	IT	5	-10.0	85.0
25	126	Neha	Female	22.0	ECE	6	93.0	101.0
26	127	Rohit	Male	20.0	CSE	5	67.0	75.0



5. Heatmap

A **heatmap** visualizes data values using colors. It is commonly used to display a **correlation matrix**.

```
# Heatmap

import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd
data = pd.read_csv("student_data.csv")
df = pd.DataFrame(data)
print(df)

groupedby = df.groupby("City").agg({"Marks" : "sum"})
print(groupedby)
sns.heatmap(groupedby)
plt.show()
```

	Student_ID	Name	Gender	Age	Department	Semester	Marks	Attendance	\
0	101	Aarav	Male	20.0	CSE	5	85.0	92.0	
1	102	Priya	Female	21.0	IT	5	91.0	95.0	
2	103	Rahul	Male	22.0	ECE	6	76.0	88.0	
3	104	Anjali	Female	20.0	CSE	5	89.0	91.0	
4	105	Karan	Male	21.0	IT	5	NaN	85.0	
5	106	Neha	Female	22.0	ECE	6	93.0	NaN	
6	107	Rohit	Male	20.0	CSE	5	67.0	75.0	
7	108	Simran	Female	21.0	IT	5	82.0	89.0	
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17	118	Pooja	Female	21.0	ECE	5	87.0	90.0	
18	119	Deepak	Male	22.0	IT	6	74.0	80.0	
19	120	Sneha	Female	20.0	CSE	5	92.0	97.0	
20	121	Aarav	Male	20.0	CSE	5	85.0	92.0	
21	122	Priya	Female	21.0	IT	5	91.0	95.0	
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